Introduction

Deep learning has transformed many fields (LeCun et al., 2015). Hinton and Salakhutdinov (2006) showed that deep autoencoders can reduce dimensionality. Decisions under risk are well described by prospect theory (Kahneman & Tversky, 1979; Smith, 2021).

The transformer architecture (Vaswani et al., 2018) replaced recurrence. However, Silver et al. (2016) combined tree search with deep networks, building on convolutional models (e.g., Krizhevski et al., 2012; see also LeCun et al., 2015, p. 437).

王小明（２０２０）指出深度學習可應用於教育現場，其他研究也有類似發現（李大華、陳美玲，２０１９；張三，２０１８）。A total of 2020 participants (N = 2020) were recruited in May (
May 2019).

References

Chen, L., & Wong, T. (2022). A fabricated study that does not exist anywhere. Journal of Imaginary Results, 5(1), 1–10. https://doi.org/10.1234/fake.2022.99999

Hinton, G. E., & Salakhutdinov, R. R. (2006). Reducing the dimensionality of data with neural networks. Science, 313(5786), 504–507. https://doi.org/10.1126/science.1127647

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. Econometrica, 47(2), 263–291.

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). ImageNet classification with deep convolutional neural networks. Advances in Neural Information Processing Systems, 25, 1097–1105.

LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. Nature, 521(7553), 436–444. https://doi.org/10.1038/nature14539

Silver, D., Huang, A., Maddison, C. J., Guez, A., Sifre, L., & Hassabis, D. (2015). Mastering the game of Go with deep neural networks and tree search. Nature, 529(7587), 484–489. https://doi.org/10.1038/nature16961

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. Advances in Neural Information Processing Systems, 30, 5998–6008.

Watson, J. D., & Crick, F. H. C. (1953). Molecular structure of nucleic acids: A structure for deoxyribose nucleic acid. Nature, 171(4356), 737–738. https://doi.org/10.1038/nature14539

王小明（２０２０）。深度學習在教育上的應用。教育研究期刊，１２（３），４５－６７.

李大華、陳美玲（２０１９）。人工智慧與課程設計。課程與教學季刊，２２（１），１－30.